

CS PhD Seminar Series

October 13th

| 14:30-15:30

| Room 322

Finite State Controllers for Model-free Olfactory Navigation

Olfactory navigation is a fundamental problem faced by many organisms. Because odor signals are sparse, intermittent, and noisy, this task can be naturally formulated as a Partially Observable Markov Decision Process (POMDP), which can be approximately solved with Reinforcement Learning (RL) techniques. Despite the Markov property of the state of the process, partial observability restricts the information an agent obtains, hence keeping track of previous observations and action is fundamental to act optimally. Finite State Controllers (FSC) are one way to represent an agent with a notion of memory that, much like neural networks, learns what is worthwhile to remember. On the other hand, FSC operates under heavy constraint on the dimensionality of the memory, making them less computationally cumbersome and more explainable. In my work, I show that it is possible to learn an optimal FSC with minimal prior knowledge of the environment and that they can adapt to different odor landscapes.



Speaker: **Andrea Marchi**

Andrea Marchi is a first-year PhD student in Computer Science at University of Genova, holding an MSc in Data Science and Artificial Intelligence from the University of Genova. His work focuses on model-free reinforcement learning applied to olfactory navigation and understanding decisions made by biological organisms, in particular how to balance optimality of decisions with computational and memory constraints.

Where the Training Signal Comes From: Label Generation for Face Parsing and Reward-Based Sketch Completion

Face parsing degrades at extreme poses due to frontal-heavy benchmarks and tedious manual labelling. We solve this by fitting two 3D Gaussian Splatting models to multi-view captures: one to photos and another to baseline masks, using shared geometry to eliminate view-specific errors and yield perfectly aligned synthetic labels. Fine-tuning on such pairs outperform existing parsers without loss of frontal accuracy. To tackle incomplete forensic sketches where witnesses recall only select features, traditional pixel-matching reconstruction losses fail because they miss crucial identity traits. Instead, we fine-tune a diffusion inpainter using a sketch-adapted face embedder as a direct reward signal. The resulting completions reliably move closer to true identities, transfer seamlessly to unseen hand-drawn sketches.

Speaker: **Ankit Gahlawat**

Ankit Gahlawat is a PhD researcher in Artificial Intelligence at IIT Bangalore, working in the Multimodal Perception Lab under Prof. Dinesh Babu Jayagopi and Prof. Viswanath Gopalakrishnan. His research focuses on computer vision, generative models and 3D Gaussian Splatting, with an emphasis on robust facial reconstruction, identity-preserving synthesis under extreme poses and severe occlusions and reward hacking. Before transitioning to research, he worked in software engineering roles, building scalable backend systems, recommendation engines and data processing platforms.

